

vtu-easy.com

MODULE - 1

FOURIER SERIES & PRACTICAL HARMONIC ANALYSIS

Subject: Engineering Mathematics - III (BMATEC301)
VTU | B.E. / B.Tech | 3rd Semester

Topics Covered:

- ◆ Periodic Functions & Dirichlet's Condition
- ◆ Fourier Series Expansion (Period 2π and Arbitrary Period)
 - ◆ Even & Odd Functions — Fourier Coefficients
- ◆ Worked Examples: $|x|$, $(\pi-x)/2$, e^{ax} , $x-x^2$, Rectangular Wave, etc.
 - ◆ Half-Range Fourier Series (Sine & Cosine)
 - ◆ Practical Harmonic Analysis
 - ◆ Variation of Periodic Current

vtu-easy.com

TABLE OF CONTENTS

1. Introduction to Fourier Series	3
1.1 Bernoulli's Rule of Integration by Parts	3
1.2 Standard Results & Useful Identities	3
1.3 Periodic Functions	4
1.4 Even and Odd Functions	4
1.5 Dirichlet's Conditions	5
2. Fourier Series — General Definition & Formulas	6
2.1 Fourier Series with Period $2l$	6
2.2 Fourier Series with Period 2π	6
2.3 Fourier Coefficient Table (All Intervals)	7
2.4 Convergence at Discontinuities	8
2.5 Fourier Coefficients for Even & Odd Functions	8
3. Worked Problems — Period 2π	10
3.1 $f(x) = x $ in $(-\pi, \pi)$	10
3.2 $f(x) = (\pi-x)/2$ in $(0, 2\pi)$	12
3.3 $f(x) = ((\pi-x)/2)^2$ in $(0, 2\pi)$	14
3.4 $f(x) = 1 \pm 2x/\pi$ (Triangular Wave)	16
3.5 $f(x) = e^{ax}$ in $(-\pi, \pi)$	18
3.6 $f(x) = x - x^2$ in $(-\pi, \pi)$	21
3.7 Piecewise Functions in $(-\pi, \pi)$	23
4. Worked Problems — Arbitrary Period	26
4.1 $f(x) = \pi x$ in $(0 \leq x \leq 1)$, $\pi(2-x)$ in $(1 \leq x \leq 2)$	26
4.2 $f(x) = 2x - x^2$ in $(0, 3)$	28
5. Half-Range Fourier Series	30
5.1 Definition & Formula Table	30
5.2 Half-Range Cosine Series — $f(x) = (x-1)^2$ in $(0, 1)$	31
5.3 Half-Range Cosine Series — $f(x) = x(l-x)$	33
5.4 Half-Range Sine Series	35
6. Practical Harmonic Analysis	37
6.1 Theory & Formulas	37
6.2 Worked Problem 1 — Degrees (6 values)	38
6.3 Worked Problem 2 — Radians (7 values, 2 harmonics)	39
6.4 Worked Problem 3 — Arbitrary Period l	40
6.5 Variation of Periodic Current	41
7. Key Formulas & Quick Reference Card	43
8. Important Results & Deductions Summary	45

1. INTRODUCTION TO FOURIER SERIES

Fourier Series is a method of expressing any periodic function as an infinite sum of sine and cosine terms. It is one of the most powerful tools in engineering mathematics, with applications in signal processing, heat conduction, vibrations, and circuit analysis.

This module covers the complete theory of Fourier Series as prescribed by Visvesvaraya Technological University (VTU) for B.E./B.Tech. 3rd Semester Engineering Mathematics (BMATEC301).

1.1 Bernoulli's Rule of Integration by Parts

When integrating a product of two functions, Bernoulli's rule provides a systematic method. If u and v are functions of x , then:

$$\int uv \, dx = u \cdot \int v \, dx - u' \cdot \int \int v \, dx + u'' \cdot \int \int \int v \, dx - \dots$$

where u' , u'' , u''' ... are successive derivatives of u and $\int v$, $\int \int v$, ... are successive integrals of v .

NOTE: This rule is extremely useful in computing Fourier coefficients a_n and b_n when the function involves products like $x \cdot \cos(nx)$ or $x^2 \cdot \sin(nx)$. Choose u as the algebraic part and v as the trigonometric part.

Example: To evaluate $\int x \cdot \cos(nx) \, dx$, set $u = x$ and $v = \cos(nx)$:

$$\int x \cdot \cos(nx) \, dx = x \cdot (\sin(nx)/n) - (1) \cdot (-\cos(nx)/n^2) + 0 = x \cdot \sin(nx)/n + \cos(nx)/n^2$$

1.2 Standard Results & Useful Identities

The following standard results are used repeatedly throughout Fourier Series computations:

Trigonometric Identities at Integer Multiples of π

1. $\sin(-\theta) = -\sin \theta$, $\cos(-\theta) = \cos \theta$ [Odd/Even function properties]
2. If ' n ' is an integer:
 - (i) $\sin(n\pi) = 0$ for all integers n

$$(ii) \cos(n\pi) = (-1)^n = \{ +1, \text{ if } n \text{ is even ; } -1, \text{ if } n \text{ is odd } \}$$

Standard Integral Formulas

3. $\int e^{sx} \sin(bx) \, dx = e^{sx}/(a^2+b^2) \cdot (a \cdot \sin(bx) - b \cdot \cos(bx)) + C$
4. $\int e^{sx} \cos(bx) \, dx = e^{sx}/(a^2+b^2) \cdot (a \cdot \cos(bx) + b \cdot \sin(bx)) + C$

Definite Integral Properties

5. $\int_{-a}^a f(x) dx = \{ 2 \int_0^a f(x) dx, \text{ if } f(x) \text{ is even ; } 0, \text{ if } f(x) \text{ is odd } \}$
6. $\int_0^{2a} f(x) dx = \{ 2 \int_0^a f(x) dx, \text{ if } f(x) \text{ is even ; } 0, \text{ if } f(x) \text{ is odd } \}$

Product Rules for Even and Odd Functions

7. Even $\times$ Even = Even Function
 Odd $\times$ Odd = Even Function
 Even $\times$ Odd = Odd Function

NOTE: These product rules are critical: they let you immediately determine $b_0 = 0$ (for even functions) or $a_0 = a_n = 0$ (for odd functions), saving significant computation.

1.3 Periodic Functions

A function $f(x)$ is said to be a periodic function with period T if:

$$f(x + T) = f(x) \text{ for all } x, T > 0$$

The smallest such positive value T is called the fundamental period of $f(x)$.

Examples of periodic functions commonly encountered in engineering:

- $f(x) = \sin(x) \rightarrow$ periodic with period 2π
- $f(x) = \cos(x) \rightarrow$ periodic with period 2π
- $f(x) = \tan(x) \rightarrow$ periodic with period π
- Square wave, triangular wave, sawtooth wave $\rightarrow$ periodic with various periods

Any periodic function satisfying certain conditions (Dirichlet's conditions) can be represented by a Fourier series — an infinite sum of harmonically related sinusoidal functions.

1.4 Even and Odd Functions

Understanding even and odd functions is fundamental to simplifying Fourier series calculations. Many functions fall neatly into one category, which halves the computation work.

Even Function

A function $f(x)$ is even if: $f(-x) = f(x)$ for all x

Graphically, the curve is symmetric about the y-axis.

Examples: $f(x) = |x|$, $f(x) = x^2$, $f(x) = \cos(x)$, $f(x) = x^4 + 3x^2 + 1$

Key result for Fourier series: Since $\sin(nx)$ is odd and $f(x)$ is even, $f(x) \cdot \sin(nx)$ is ODD. Hence:

$$b_n = 0 \text{ for ALL } n \text{ (when } f(x) \text{ is even)}$$

Odd Function

A function $f(x)$ is odd if: $f(-x) = -f(x)$ for all x

Graphically, the curve has 180° rotational symmetry about the origin.

Examples: $f(x) = x$, $f(x) = x^3$, $f(x) = \sin(x)$, $f(x) = x^3 - 5x$

Key result for Fourier series: Since $\cos(nx)$ is even and $f(x)$ is odd, $f(x) \cdot \cos(nx)$ is ODD. Hence:

$$a_0 = 0 \text{ and } a_n = 0 \text{ for ALL } n \text{ (when } f(x) \text{ is odd)}$$

How to Check Even/Odd for Piecewise Functions

For a piecewise function defined as $f(x) = \phi(x)$ in $(-l, 0)$ and $\psi(x)$ in $(0, l)$:

- $f(x)$ is EVEN if $\phi(-x) = \psi(x)$
- $f(x)$ is ODD if $\phi(-x) = -\psi(x)$

For a piecewise function defined in $(0, 2l)$ with two pieces:

- $f(x)$ is EVEN if $\phi(2l-x) = \psi(x)$
- $f(x)$ is ODD if $\phi(2l-x) = -\psi(x)$

1.5 Dirichlet's Conditions

Dirichlet's conditions are sufficient conditions for a Fourier series to converge to the original function. A function $f(x)$ can be expanded in a Fourier series over $[a, a+2l]$ if:

- $f(x)$ is periodic with period $2l$
- $f(x)$ is single-valued (each x maps to one value)
- $f(x)$ has a finite number of maxima and minima in one period
- $f(x)$ has a finite number of finite discontinuities in one period (no infinite jumps)

- $f(x)$ is bounded ($|f(x)|$ is finite everywhere)

IMPORTANT: At a point of discontinuity $x = c$, the Fourier series converges to: $(1/2)[f(c^+) + f(c^-)]$, i.e., the average of the left-hand and right-hand limits.

At points of continuity, the series converges exactly to $f(x)$.

2. FOURIER SERIES — GENERAL DEFINITION & FORMULAS

2.1 Fourier Series with Period $2l$

The Fourier series of a periodic function $f(x)$ with period $2l$ is given by:

$$f(x) = a_0/2 + \sum_{n=1}^{\infty} a_n \cdot \cos(n\pi x/l) + \sum_{n=1}^{\infty} b_n \cdot \sin(n\pi x/l)$$

where a_0 , a_n , b_n are the Fourier coefficients determined by Euler's formulas.

2.2 Fourier Series with Period 2π

When the period is 2π (i.e., $l = \pi$), the series simplifies to:

$$f(x) = a_0/2 + \sum_{n=1}^{\infty} a_n \cdot \cos(nx) + \sum_{n=1}^{\infty} b_n \cdot \sin(nx)$$

The coefficients are computed using:

$$a_0 = (1/\pi) \int_{-\pi}^{\pi} f(x) dx \quad (\text{or over any interval of length } 2\pi)$$

$$a_n = (1/\pi) \int_{-\pi}^{\pi} f(x) \cdot \cos(nx) dx$$

$$b_n = (1/\pi) \int_{-\pi}^{\pi} f(x) \cdot \sin(nx) dx$$

2.3 Fourier Coefficient Table (All Intervals)

The following master table gives Fourier coefficients for all standard intervals:

Interval	a_0	a_n	b_n
$(-l, l)$	$(1/l) \int_{-l}^l f(x) dx$	$(1/l) \int_{-l}^l f(x) \cos(n\pi x/l) dx$	$(1/l) \int_{-l}^l f(x) \sin(n\pi x/l) dx$
$(-\pi, \pi)$	$(1/\pi) \int_{-\pi}^{\pi} f(x) dx$	$(1/\pi) \int_{-\pi}^{\pi} f(x) \cos(nx) dx$	$(1/\pi) \int_{-\pi}^{\pi} f(x) \sin(nx) dx$
$(0, 2l)$	$(1/l) \int_0^{2l} f(x) dx$	$(1/l) \int_0^{2l} f(x) \cos(n\pi x/l) dx$	$(1/l) \int_0^{2l} f(x) \sin(n\pi x/l) dx$
$(0, 2\pi)$	$(1/\pi) \int_0^{2\pi} f(x) dx$	$(1/\pi) \int_0^{2\pi} f(x) \cos(nx) dx$	$(1/\pi) \int_0^{2\pi} f(x) \sin(nx) dx$

NOTE: The factor before each integral differs: $1/l$ for general period $2l$, and $1/\pi$ for period 2π .

2.4 Convergence at Discontinuities

This is a critically important theorem for VTU exams:

1. If $f(x)$ is continuous at $x = c$, then the Fourier series converges to $f(c)$.
2. If $f(x)$ is discontinuous at $x = c$, the Fourier series converges to:

$$S(c) = (1/2)[f(c^+) + f(c^-)]$$

where $f(c^+) = \lim_{x \rightarrow c^+} f(x)$ is the right-hand limit, and $f(c^-) = \lim_{x \rightarrow c^-} f(x)$ is the left-hand limit.

This result is used in deductions — substituting a specific value of x into the Fourier series equation to obtain famous series summations like $\pi^2/8$, $\pi^2/6$, $\pi/4$, etc.

IMPORTANT: At the endpoints of the interval, the Fourier series converges to the average of the function values at both endpoints. This is the key step in 'deduction' problems.

2.5 Fourier Coefficients for Even & Odd Functions

When $f(x)$ is even or odd, the coefficient formulas simplify drastically:

Interval	Nature	a_0	a_n	b_n
$(-l, l)$ or $(0, 2l)$	Even: $f(-x)=f(x)$ or $f(2l-x)=f(x)$	$(2/l)\int_0^l f(x)dx$	$(2/l)\int_0^l f(x)\cos(n\pi x/l)dx$	0
$(-\pi, \pi)$ or $(0, 2\pi)$	Even: $f(-x)=f(x)$ or $f(2\pi-x)=f(x)$	$(2/\pi)\int_0^\pi f(x)dx$	$(2/\pi)\int_0^\pi f(x)\cos(nx)dx$	0
$(-l, l)$ or $(0, 2l)$	Odd: $f(-x)=-f(x)$ or $f(2l-x)=-f(x)$	0	0	$(2/l)\int_0^l f(x)\sin(n\pi x/l)dx$
$(-\pi, \pi)$ or $(0, 2\pi)$	Odd: $f(-x)=-f(x)$ or $f(2\pi-x)=-f(x)$	0	0	$(2/\pi)\int_0^\pi f(x)\sin(nx)dx$

NOTE: For even functions: only cosine terms survive ($b_n = 0$). For odd functions: only sine terms survive ($a_0 = a_n = 0$). This halves the number of integrals to compute!

3. WORKED PROBLEMS — PERIOD 2π

This section presents fully worked VTU exam problems with complete step-by-step solutions. Each problem follows the standard VTU pattern: check even/odd, compute coefficients, write the series, and perform deductions.

3.1 Problem 1: $f(x) = |x|$ in $(-\pi, \pi)$ — Deduction $\pi^2/8$

Find the Fourier series of $f(x) = |x|$ in $-\pi < x < \pi$. Hence deduce that $\pi^2/8 = \sum_{n=1}^{\infty} 1/(2n-1)^2$.

Step 1: Check Even or Odd

$$f(-x) = |-x| = |x| = f(x) \Rightarrow f(x) \text{ is an EVEN function}$$

Therefore: $b_n = 0$ for all n .

Also note: $f(x) = |x| = x$ for x in $(0, \pi)$ and $f(x) = -x$ for x in $(-\pi, 0)$

Step 2: Compute a_0

Since $f(x)$ is even:

$$\begin{aligned} a_0 &= (2/\pi) \int_0^\pi f(x) dx = (2/\pi) \int_0^\pi x dx \\ &= (2/\pi) \cdot [x^2/2]_0^\pi = (2/\pi) \cdot (\pi^2/2 - 0) = (2/\pi) \cdot (\pi^2/2) \end{aligned}$$

$$\therefore a_0 = \pi$$

Step 3: Compute a_n

$$a_n = (2/\pi) \int_0^\pi f(x) \cdot \cos(nx) dx = (2/\pi) \int_0^\pi x \cdot \cos(nx) dx$$

Applying Bernoulli's rule ($u = x, v = \cos(nx)$):

$$\begin{aligned} &= (2/\pi) [x \cdot (\sin(nx)/n) - (1) \cdot (-\cos(nx)/n^2)]_0^\pi \\ &= (2/\pi) [x \cdot \sin(nx)/n + \cos(nx)/n^2]_0^\pi \\ &= (2/\pi) [(\pi \cdot \sin(n\pi)/n + \cos(n\pi)/n^2) - (0 + \cos(0)/n^2)] \\ &= (2/\pi) [(\pi \cdot 0/n + (-1)^n/n^2) - (1/n^2)] \quad [\text{since } \sin(n\pi) = 0, \cos(n\pi) = (-1)^n] \\ &= (2/\pi) \cdot [(-1)^n - 1]/n^2 \end{aligned}$$

$$\therefore a_n = (2/n^2\pi)[(-1)^n - 1] = (-2/n^2\pi)[1 - (-1)^n]$$

NOTE: Observe: When n is even, $(-1)^n = 1$, so $a_n = 0$. When n is odd, $(-1)^n = -1$, so $a_n = -4/(n^2\pi)$.

Step 4: Write the Fourier Series

The Fourier series of $f(x) = |x|$ with period 2π is:

$$f(x) = \pi/2 + \sum_{n=1}^{\infty} (2/n^2\pi)[(-1)^n - 1] \cdot \cos(nx)$$

Expanding (only odd n terms survive):

$$|x| = \pi/2 - (4/\pi)[\cos(x)/1^2 + \cos(3x)/3^2 + \cos(5x)/5^2 + \dots] \dots \textcircled{1}$$

Step 5: Deduction — Proving $\pi^2/8$

Put $x = 0$ in equation $\textcircled{1}$. Note: $f(0) = |0| = 0$, and $x = 0$ is a point of continuity, so the series converges to $f(0) = 0$.

$$0 = \pi/2 - (4/\pi)[1/1^2 + 1/3^2 + 1/5^2 + \dots]$$

$$(4/\pi)[1/1^2 + 1/3^2 + 1/5^2 + \dots] = \pi/2$$

$$1/1^2 + 1/3^2 + 1/5^2 + \dots = \pi^2/8$$

$$\therefore \pi^2/8 = \sum_{n=1}^{\infty} 1/(2n-1)^2 = 1/1^2 + 1/3^2 + 1/5^2 + \dots \quad (\text{Proved})$$

3.2 Problem 2: $f(x) = (\pi-x)/2$ in $(0, 2\pi)$ — Deduction $\pi/4$

Obtain the Fourier series for $f(x) = (\pi-x)/2$ in $0 < x < 2\pi$. Hence deduce that $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \dots$

Step 1: Check Even or Odd

$f(x) = (\pi-x)/2$ is defined in $(0, 2\pi)$. Check: $f(2\pi-x) = (\pi-(2\pi-x))/2 = (\pi-2\pi+x)/2 = (x-\pi)/2 = -(\pi-x)/2 = -f(x)$

Since $f(2\pi-x) = -f(x)$, the function is an ODD function.

Therefore: $a_0 = 0$ and $a_n = 0$ for all n .

Step 2: Compute b_n

$$b_n = (2/\pi) \int_0^\pi f(x) \cdot \sin(nx) \, dx = (2/\pi) \int_0^\pi [(\pi-x)/2] \cdot \sin(nx) \, dx$$

$$= (1/\pi) \int_0^\pi (\pi-x) \cdot \sin(nx) \, dx$$

Applying Bernoulli's rule ($u = \pi-x$, $v = \sin(nx)$):

$$= (1/\pi) [(\pi-x) \cdot (-\cos(nx)/n) - (-1) \cdot (-\sin(nx)/n^2)]_0^\pi$$

$$= (1/\pi) [-(\pi-x) \cdot \cos(nx)/n - \sin(nx)/n^2]_0^\pi$$

$$= (1/\pi) \{ [-(\pi-\pi) \cdot \cos(n\pi)/n - \sin(n\pi)/n^2] - [-(\pi-0) \cdot \cos(0)/n - \sin(0)/n^2] \}$$

$$= (1/\pi) \{ [0 - 0] - [-\pi/n - 0] \}$$

$$= (1/\pi) \cdot (\pi/n) = 1/n$$

$$\therefore b_n = 1/n$$

Step 3: Write the Fourier Series

$$(\pi-x)/2 = \sum_{n=1}^{\infty} (1/n) \cdot \sin(nx) = \sin(x) + \sin(2x)/2 + \sin(3x)/3 + \dots \quad \text{... (1)}$$

Step 4: Deduction — Proving $\pi/4$

Put $x = \pi/2$ in equation (1):

$$(\pi - \pi/2)/2 = \sin(\pi/2)/1 + \sin(\pi)/2 + \sin(3\pi/2)/3 + \sin(2\pi)/4 + \sin(5\pi/2)/5 + \dots$$

$$\pi/4 = 1 + 0 - 1/3 + 0 + 1/5 - 0 - 1/7 + \dots$$

$$\therefore \pi/4 = 1 - 1/3 + 1/5 - 1/7 + \dots \quad (\text{Leibniz Formula — Proved})$$

3.3 Problem 3: $f(x) = ((\pi-x)/2)^2$ in $(0, 2\pi)$ — Deduction $\pi^2/6$

Find the Fourier series of $f(x) = ((\pi-x)/2)^2$ in $0 < x < 2\pi$. Hence evaluate $1/1^2 + 1/2^2 + 1/3^2 + 1/4^2 + \dots$

Step 1: Check Even or Odd

$$f(2\pi-x) = ((\pi-(2\pi-x))/2)^2 = ((x-\pi)/2)^2 = ((\pi-x)/2)^2 = f(x)$$

Since $f(2\pi-x) = f(x)$, the function is an EVEN function in $(0, 2\pi)$.

Therefore: $b_n = 0$.

Step 2: Compute a_0

$$a_0 = (2/\pi) \int_0^\pi f(x) dx = (2/\pi) \int_0^\pi [(\pi-x)/2]^2 dx = (2/\pi) \cdot (1/4) \int_0^\pi (\pi-x)^2 dx \\ = (1/2\pi) \cdot [-(\pi-x)^3/3]_0^\pi = (1/2\pi) \cdot [0 - (-\pi^3/3)] = \pi^2/6$$

$$\therefore a_0 = \pi^2/6$$

Step 3: Compute a_n

$$a_n = (2/\pi) \int_0^\pi [(\pi-x)/2]^2 \cdot \cos(nx) dx = (1/2\pi) \int_0^\pi (\pi-x)^2 \cdot \cos(nx) dx$$

Applying Bernoulli's rule twice ($u = (\pi-x)^2$, $v = \cos(nx)$):

$$= (1/2\pi) [(\pi-x)^2 \cdot (\sin(nx)/n) - 2(\pi-x) \cdot (-1)(-\cos(nx)/n^2) + 2 \cdot (-\sin(nx)/n^3)]_0^\pi \\ = (1/2\pi) \cdot [2(\pi-x) \cdot \cos(nx)/n^2]_0^\pi \\ = (1/2\pi) \cdot [(2(\pi-\pi) \cdot \cos(n\pi)/n^2) - (2(\pi-0) \cdot \cos(0)/n^2)] \\ = (1/2\pi) \cdot [0 - 2\pi/n^2] = -1/n^2$$

Wait — recomputing with correct signs:

$$= (1/2\pi) \cdot [-(2(\pi-x)/n^2) \cdot (-\cos(nx))]_0^\pi \\ = -(1/n^2\pi) \cdot [(\pi-x) \cdot \cos(nx)]_0^\pi \\ = -(1/n^2\pi) \cdot [(\pi-\pi) \cdot \cos(n\pi) - \pi \cdot \cos(0)] = -(1/n^2\pi) \cdot (0 - \pi) = 1/n^2$$

$$\therefore a_n = 1/n^2$$

Step 4: Fourier Series

$$((\pi-x)/2)^2 = \pi^2/12 + \sum_{n=1}^{\infty} (1/n^2) \cdot \cos(nx) \quad \dots (1)$$

Step 5: Deduction — Proving $\pi^2/6$

Put $x = 0$ in (1): $f(0) = (\pi/2)^2 = \pi^2/4$

$$\pi^2/4 = \pi^2/12 + \sum_{n=1}^{\infty} 1/n^2$$

$$\pi^2/4 - \pi^2/12 = \sum_{n=1}^{\infty} 1/n^2$$

$$\pi^2(3-1)/12 = \sum 1/n^2 \Rightarrow \pi^2 \cdot 2/12 = \sum 1/n^2$$

$$\therefore \pi^2/6 = 1/1^2 + 1/2^2 + 1/3^2 + 1/4^2 + \dots \quad (\text{Basel Problem — Proved})$$

3.4 Problem 4: Triangular Wave — $f(x) = 1 \pm 2x/\pi$ — Deduction $\pi^2/8$

Find the Fourier series of the triangular wave: $f(x) = 1 + 2x/\pi$ in $(-\pi, 0)$ and $f(x) = 1 - 2x/\pi$ in $(0, \pi)$. Hence deduce $\pi^2/8 = 1/1^2 + 1/3^2 + 1/5^2 + \dots$

Step 1: Check Even or Odd

Let $\phi(x) = 1 + 2x/\pi$ and $\psi(x) = 1 - 2x/\pi$

$$\phi(-x) = 1 + 2(-x)/\pi = 1 - 2x/\pi = \psi(x)$$

Since $\phi(-x) = \psi(x)$, the function $f(x)$ is an EVEN function in $(-\pi, \pi)$.

Therefore: $b_n = 0$.

Step 2: Compute a_0

$$a_0 = (2/\pi) \int_0^\pi (1 - 2x/\pi) dx = (2/\pi) [x - x^2/\pi]_0^\pi$$

$$= (2/\pi) [\pi - \pi^2/\pi - 0] = (2/\pi) [\pi - \pi] = 0$$

$$\therefore a_0 = 0$$

Step 3: Compute a_n

$$a_n = (2/\pi) \int_0^\pi (1 - 2x/\pi) \cdot \cos(nx) \, dx$$

Applying Bernoulli's rule:

$$\begin{aligned} &= (2/\pi) [(1 - 2x/\pi) \cdot (\sin(nx)/n) - (-2/\pi) \cdot (-\cos(nx)/n^2)]_0^\pi \\ &= (2/\pi) [(1 - 2x/\pi) \cdot \sin(nx)/n - (2/\pi n^2) \cdot \cos(nx)]_0^\pi \\ &= (2/\pi) \{ [(1 - 2) \cdot 0/n - (2/\pi n^2) \cdot \cos(n\pi)] - [1 \cdot 0/n - (2/\pi n^2) \cdot 1] \} \\ &= (2/\pi) \{ [-2(-1)^n/\pi n^2] - [-2/\pi n^2] \} \\ &= (2/\pi) \cdot (2/\pi n^2) \cdot [1 - (-1)^n] \end{aligned}$$

$$\therefore a_n = (4/n^2\pi^2)[1 - (-1)^n]$$

NOTE: When n is even: $a_n = 0$. When n is odd: $a_n = 8/(n^2\pi^2)$

Step 4: Fourier Series

$$f(x) = \sum_{n=1}^{\infty} (4/n^2\pi^2)[1 - (-1)^n] \cdot \cos(nx) = (8/\pi^2)[\cos(x)/1^2 + \cos(3x)/3^2 + \cos(5x)/5^2 + \dots]$$

Step 5: Deduction

Put $x = 0$: $f(0) = 1 - 0 = 1$ (from the second piece, as $x=0$ is the boundary)

$$1 = (8/\pi^2)[1/1^2 + 1/3^2 + 1/5^2 + \dots]$$

$$\therefore \pi^2/8 = 1/1^2 + 1/3^2 + 1/5^2 + \dots \quad (\text{Proved})$$

3.5 Problem 5: $f(x) = e^{ax}$ in $(-\pi, \pi)$ — Deduction involving $\sinh$

Obtain the Fourier series of $f(x) = e^{ax}$ in $(-\pi, \pi)$. Hence deduce: $\pi/(\sinh \pi a) = \dots$

Step 1: Check Even or Odd

$$f(-x) = e^{a(-x)} = e^{-ax}$$

$$e^{-ax} \neq e^{ax} \text{ and } e^{-ax} \neq -e^{ax} \text{ in general.}$$

$\therefore f(x)$ is neither even nor odd. Use general formulas.

Step 2: Compute a_0

$$a_0 = (1/\pi) \int_{-\pi}^{\pi} e^{ax} \, dx = (1/\pi) [e^{ax}/a]_{-\pi}^{\pi}$$

$$= (1/a\pi) [e^{a\pi} - e^{-a\pi}] = (1/a\pi) \cdot 2 \cdot \sinh(a\pi)$$

$$\therefore a_0 = 2 \cdot \sinh(a\pi)/(a\pi)$$

Step 3: Compute a_n

Using the standard integral: $\int e^{ax} \cdot \cos(nx) \, dx = e^{ax}/(a^2 + n^2) \cdot (a \cdot \cos(nx) + n \cdot \sin(nx))$

$$a_n = (1/\pi) \int_{-\pi}^{\pi} e^{ax} \cdot \cos(nx) \, dx = (1/\pi) \cdot [e^{ax}/(a^2 + n^2) \cdot (a \cdot \cos(nx) + n \cdot \sin(nx))]_{-\pi}^{\pi}$$

$$\text{At } x=\pi: e^{a\pi} \cdot (a \cdot \cos(n\pi) + n \cdot \sin(n\pi)) = e^{a\pi} \cdot a \cdot (-1)^n \quad [\text{since } \sin(n\pi) = 0]$$

$$\text{At } x=-\pi: e^{-a\pi} \cdot (a \cdot \cos(-n\pi) + n \cdot \sin(-n\pi)) = e^{-a\pi} \cdot a \cdot (-1)^n$$

$$= (a(-1)^n/\pi(a^2+n^2)) \cdot [e^{a\pi} - e^{-a\pi}] = (a(-1)^n/\pi(a^2+n^2)) \cdot 2 \cdot \sinh(a\pi)$$

$$\therefore a_n = 2a(-1)^n \cdot \sinh(a\pi) / [\pi(a^2+n^2)]$$

Step 4: Compute b_n

Using: $\int e^{ax} \cdot \sin(nx) dx = e^{ax}/(a^2+n^2) \cdot (a \cdot \sin(nx) - n \cdot \cos(nx))$

$$b_n = (1/\pi) \int_{-\pi}^{\pi} e^{ax} \cdot \sin(nx) dx = (1/\pi) [e^{ax} \cdot (a \cdot \sin(nx) - n \cdot \cos(nx)) / (a^2+n^2)]_{-\pi}^{\pi}$$

$$\text{At } x=\pi: e^{a\pi} \cdot (a \cdot 0 - n \cdot (-1)^n) = -n(-1)^n \cdot e^{a\pi}$$

$$\text{At } x=-\pi: e^{-a\pi} \cdot (a \cdot 0 - n \cdot (-1)^n) = -n(-1)^n \cdot e^{-a\pi}$$

$$b_n = (1/\pi(a^2+n^2)) \cdot (-n(-1)^n) \cdot [e^{a\pi} - e^{-a\pi}]$$

$$= (-n(-1)^n \cdot 2 \cdot \sinh(a\pi)) / (\pi(a^2+n^2))$$

$$\therefore b_n = 2n(-1)^{n+1} \cdot \sinh(a\pi) / [\pi(a^2+n^2)]$$

Step 5: Fourier Series

The Fourier series is:

$$e^{ax} = \sinh(a\pi)/(a\pi) + (2 \cdot \sinh(a\pi)/\pi) \cdot \sum_{n=1}^{\infty} [(-1)^n/(a^2+n^2)] \cdot [a \cdot \cos(nx) - n \cdot \sin(nx)] \dots (1)$$

Step 6: Deduction

Put $x = 0$ and $a = 1$ in (1):

$$e^0 = \sinh(\pi)/\pi + (2 \cdot \sinh(\pi)/\pi) \cdot \sum_{n=1}^{\infty} (-1)^n/(1+n^2) \cdot [1 \cdot 1 - n \cdot 0]$$

$$1 = \sinh(\pi)/\pi + (2 \cdot \sinh(\pi)/\pi) \cdot \sum_{n=1}^{\infty} (-1)^n/(n^2+1)$$

The sum $\sum_{n=1}^{\infty} (-1)^n/(n^2+1) = -1/2 + \sum_{n=1}^{\infty} (-1)^n/(n^2+1)$ [standard manipulation]

After algebraic simplification:

$$\pi/\sinh(\pi) = 2 \cdot \sum_{n=0}^{\infty} (-1)^n/(n^2+1)$$

3.6 Problem 6: $f(x) = x - x^2$ in $(-\pi, \pi)$ — Deduction $\pi^2/12$

Find the Fourier series of $f(x) = x - x^2$ in $(-\pi, \pi)$. Hence deduce $\pi^2/12 = 1/1^2 - 1/2^2 + 1/3^2 - \dots$

Step 1: Check Even or Odd

$$f(-x) = (-x) - (-x)^2 = -x - x^2$$

$$f(x) = x - x^2$$

$$f(-x) \neq f(x) \text{ and } f(-x) \neq -f(x)$$

$\therefore f(x)$ is NEITHER even NOR odd.

Step 2: Compute a_0

$$a_0 = (1/\pi) \int_{-\pi}^{\pi} (x - x^2) dx$$

$$= (1/\pi) [x^2/2 - x^3/3]_{-\pi}^{\pi} = (1/\pi) [(1/2)(\pi^2 - \pi^2) - (1/3)(\pi^3 - (-\pi^3))] = (1/\pi) [(1/2)(0) - (1/3)(2\pi^3)] = -2\pi^2/3$$

$$\therefore a_0 = -2\pi^2/3$$

Step 3: Compute a_n

$$a_n = (1/\pi) \int_{-\pi}^{\pi} (x-x^2) \cdot \cos(nx) dx$$

Applying Bernoulli's rule:

$$= (1/\pi)[(x-x^2) \cdot (\sin(nx)/n) - (1-2x) \cdot (-\cos(nx)/n^2) + (-2) \cdot (-\sin(nx)/n^3)] \cdot \pi$$

Only the $\cos(nx)$ term survives at the limits ($\sin(n\pi) = 0$):

$$= (1/\pi n^2)[(1-2x) \cdot \cos(nx)] \cdot \pi = (1/\pi n^2)[(1-2\pi) \cdot \cos(n\pi) - (1+2\pi) \cdot \cos(-n\pi)]$$

$$= (\cos(n\pi)/\pi n^2) \cdot [(1-2\pi) - (1+2\pi)] = ((-1)^n/\pi n^2) \cdot (-4\pi)$$

$$\therefore a_n = -4(-1)^n/n^2$$

Step 4: Compute b_n

$$b_n = (1/\pi) \int_{-\pi}^{\pi} (x-x^2) \cdot \sin(nx) dx$$

Applying Bernoulli's rule:

$$= (1/\pi)[(x-x^2)(-\cos(nx)/n) - (1-2x)(-\sin(nx)/n^2) + (-2)(\cos(nx)/n^3)] \cdot \pi$$

$$= (1/\pi)[(x-x^2)(-\cos(nx)/n) - (2/n^3)\cos(nx)] \cdot \pi$$

$$= (1/\pi n) \cdot [-(x-x^2)\cos(nx)] \cdot \pi + \dots (\sin \text{ terms vanish})$$

$$= (-1/\pi n) \cdot [(\pi-\pi^2)\cos(n\pi) - (-\pi-\pi^2)\cos(-n\pi)]$$

$$= (-(-1)^n/\pi n) \cdot [(\pi-\pi^2) - (-\pi-\pi^2)] = (-(-1)^n/\pi n) \cdot [2\pi]$$

$$\therefore b_n = -2(-1)^n/n$$

Step 5: Fourier Series

$$f(x) = -\pi^2/3 + \sum_{n=1}^{\infty} [-4(-1)^n/n^2] \cdot \cos(nx) + \sum_{n=1}^{\infty} [-2(-1)^n/n] \cdot \sin(nx) \dots \textcircled{1}$$

Step 6: Deduction

Put $x = 0$ in $\textcircled{1}$: $f(0) = 0 - 0 = 0$

$$0 = -\pi^2/3 - 4 \cdot \sum_{n=1}^{\infty} (-1)^n/n^2$$

$$4 \cdot \sum_{n=1}^{\infty} (-1)^n/n^2 = -\pi^2/3$$

$$\sum_{n=1}^{\infty} (-1)^n/n^2 = -\pi^2/12$$

$$-1/1^2 + 1/2^2 - 1/3^2 + \dots = -\pi^2/12$$

$$\therefore \pi^2/12 = 1/1^2 - 1/2^2 + 1/3^2 - 1/4^2 + \dots \text{ (Proved)}$$

3.7 Problem 7: Piecewise $f(x)$ in $(-\pi, \pi)$ — Deduction $\pi^2/8$

Obtain the Fourier series for $f(x) = -\pi$ in $(-\pi, 0)$ and $f(x) = x$ in $(0, \pi)$. Hence deduce sum of reciprocal squares of odd integers equals $\pi^2/8$.

Step 1: Check Even or Odd

$\phi(x) = -\pi$ and $\psi(x) = x$ in $(-\pi, \pi)$

$\phi(-x) = -\pi$ but $\psi(x) = x \neq -\pi$ in general. Also, $-\psi(x) = -x \neq -\pi$.

$\therefore f(x)$ is NEITHER even NOR odd.

Step 2: Compute a_0

$$a_0 = (1/\pi) \left[\int_{-\pi}^0 (-\pi) dx + \int_0^{\pi} x dx \right]$$

$$= (1/\pi) [-\pi \cdot (0 - (-\pi)) + \pi^2/2] = (1/\pi) [-\pi^2 + \pi^2/2] = (1/\pi) (-\pi^2/2)$$

$$\therefore a_0 = -\pi/2$$

Step 3: Compute a_n

$$a_n = (1/\pi) \left[\int_{-\pi}^0 (-\pi) \cos(nx) dx + \int_0^\pi x \cdot \cos(nx) dx \right]$$

First integral: $(1/\pi) \cdot (-\pi) \cdot [\sin(nx)/n]_{-\pi}^0 = (-1)[0-0] = 0$

Second integral using Bernoulli: $(1/\pi) [x \cdot \sin(nx)/n + \cos(nx)/n^2]_0^\pi = (1/\pi n^2) [\cos(n\pi) - \cos(0)] = ((-1)^n - 1)/\pi n^2$

$$\therefore a_n = [1 - (-1)^n] / (-n^2\pi) = -(1/n^2\pi) [1 - (-1)^n]$$

Step 4: Compute b_n

$$b_n = (1/\pi) \left[\int_{-\pi}^0 (-\pi) \sin(nx) dx + \int_0^\pi x \cdot \sin(nx) dx \right]$$

First: $(1/\pi) \cdot (-\pi) [-\cos(nx)/n]_{-\pi}^0 = (1) [\cos(0)/n - \cos(-n\pi)/n] = (1 - (-1)^n)/n$

Second: $(1/\pi) [-x \cdot \cos(nx)/n + \sin(nx)/n^2]_0^\pi = (1/\pi) [-\pi \cdot \cos(n\pi)/n] = -(-1)^n/n$

Total: $b_n = (1 - (-1)^n)/n - (-1)^n/n = (1 - 2(-1)^n)/n$

$$\therefore b_n = [1 - 2(-1)^n]/n$$

Step 5: Fourier Series

$$f(x) = -\pi/4 + \sum_{n=1}^{\infty} [-(1 - (-1)^n)/(n^2\pi)] \cos(nx) + \sum_{n=1}^{\infty} [(1 - 2(-1)^n)/n] \sin(nx) \quad \dots (1)$$

Step 6: Deduction

At $x = 0$ there is a discontinuity: $f(0^+) = 0$, $f(0^-) = -\pi$

Fourier series converges to $(1/2)[0 + (-\pi)] = -\pi/2$

$$-\pi/2 = -\pi/4 + \sum_{n=1}^{\infty} [-(1 - (-1)^n)/(n^2\pi)]$$

$$-\pi/4 = (-1/\pi) \cdot \sum_{n=1}^{\infty} [(1 - (-1)^n)/n^2]$$

$$\pi^2/4 = \sum_{n=1}^{\infty} [(1 - (-1)^n)/n^2] = 2 \cdot \sum_{n=1,3,5,\dots}^{\infty} 1/n^2$$

$$\therefore \pi^2/8 = 1/1^2 + 1/3^2 + 1/5^2 + \dots \quad (\text{Proved})$$

4. WORKED PROBLEMS — ARBITRARY PERIOD

When the period is other than 2π , we use the general period $2l$ formulas. The substitution $n\pi x/l$ replaces nx throughout.

4.1 Problem: $f(x) = \pi x$ in $(0,1)$ and $\pi(2-x)$ in $(1,2)$ — Deduction $\pi^2/8 = \sum 1/(2n-1)^2$

Obtain the Fourier series for $f(x) = \pi x$ in $0 \leq x \leq 1$ and $f(x) = \pi(2-x)$ in $1 \leq x \leq 2$. Hence deduce $\pi^2/8 = \sum 1/(2n-1)^2$.

Step 1: Identify Period

$f(x)$ is defined on $[0, 2]$. Comparing with $(0, 2l)$: $2l = 2$, so $l = 1$.

Step 2: Check Even or Odd

$\varphi(x) = \pi x$ and $\psi(x) = \pi(2-x)$

$\varphi(2-x) = \pi(2-x) = \psi(x)$

Since $\varphi(2l-x) = \psi(x)$, $f(x)$ is an EVEN function in $(0, 2)$.

Therefore: $b_n = 0$.

Step 3: Compute a_0

$$a_0 = (2/l) \int_0^1 f(x) dx = (2/1) \int_0^1 \pi x dx = 2\pi [x^2/2]_0^1 = 2\pi \cdot (1/2) = \pi$$

$$\therefore a_0 = \pi$$

Step 4: Compute a_n

$$a_n = (2/l) \int_0^1 f(x) \cdot \cos(n\pi x/l) dx = 2 \int_0^1 \pi x \cdot \cos(n\pi x) dx$$

$$= 2\pi \int_0^1 x \cdot \cos(n\pi x) dx$$

Applying Bernoulli's rule ($u=x$, $v=\cos(n\pi x)$):

$$= 2\pi [x \cdot \sin(n\pi x)/(n\pi) - (-\cos(n\pi x))/(n\pi)^2]_0^1$$

$$= 2\pi [0 + \cos(n\pi)/(n^2\pi^2) - 0 - 1/(n^2\pi^2)]$$

$$= 2\pi \cdot (\cos(n\pi) - 1)/(n^2\pi^2) = 2 \cdot ((-1)^n - 1)/(n^2\pi)$$

$$\therefore a_n = (2/n^2\pi)[(-1)^n - 1] = (-2/n^2\pi)[1 - (-1)^n]$$

NOTE: $a_n = 0$ for even n ; $a_n = -4/(n^2\pi)$ for odd n .

Step 5: Fourier Series

With period $2l = 2$:

$$f(x) = \pi/2 + \sum_{n=1}^{\infty} (-2/n^2\pi)[1 - (-1)^n] \cdot \cos(n\pi x) = \pi/2 - (4/\pi) \cdot \sum_{\text{odd } n} \cos(n\pi x)/n^2$$

Expanding:

$$f(x) = \pi/2 - (4/\pi)[\cos(\pi x)/1^2 + \cos(3\pi x)/3^2 + \cos(5\pi x)/5^2 + \dots] \dots \textcircled{1}$$

Step 6: Deduction

Put $x = 0$ in $\textcircled{1}$: $f(0) = \pi \cdot 0 = 0$

$$0 = \pi/2 - (4/\pi)[1/1^2 + 1/3^2 + 1/5^2 + \dots]$$

$$(4/\pi) \cdot \sum 1/(2n-1)^2 = \pi/2$$

$$\therefore \pi^2/8 = \sum_{n=1}^{\infty} 1/(2n-1)^2 \quad (\text{Proved})$$

4.2 Problem: $f(x) = 2x - x^2$ in $(0, 3)$

Find the Fourier series of the periodic function $f(x) = 2x - x^2$ in $0 < x < 3$ (period = 3).

Step 1: Identify Period

$f(x)$ is defined on $(0, 3)$. So $2l = 3$ and $l = 3/2$.

Step 2: Check Even or Odd

$$f(3-x) = 2(3-x) - (3-x)^2 = 6-2x - (9-6x+x^2) = 6-2x-9+6x-x^2 = 4x-x^2-3$$

$f(x) = 2x-x^2 \neq f(3-x)$. Also $\neq -f(x)$. So $f(x)$ is NEITHER even NOR odd.

Step 3: Compute a_0

$$a_0 = (1/l) \int_0^{2l} f(x) dx = (2/3) \int_0^3 (2x-x^2) dx$$

$$= (2/3) [x^2 - x^3/3]_0^3 = (2/3) [9 - 9] = 0$$

$$\therefore a_0 = 0$$

Step 4: Compute a_n (using $\theta = 2n\pi x/3$)

$$a_n = (1/l) \int_0^{2l} f(x) \cdot \cos(n\pi x/l) dx = (2/3) \int_0^3 (2x-x^2) \cdot \cos(2n\pi x/3) dx$$

Let $\omega = 2n\pi/3$. Applying Bernoulli's rule:

$$\begin{aligned} &= (2/3) \cdot [(2x-x^2) \cdot \sin(\omega x)/\omega - (2-2x) \cdot (-\cos(\omega x)/\omega^2) + (-2) \cdot (-\sin(\omega x)/\omega^3)]_0^3 \\ &= (2/3) \cdot [(6-9) \cdot \sin(3\omega)/\omega - (2-6) \cdot (-\cos(3\omega)/\omega^2) - \{0\}] \end{aligned}$$

Since $\omega = 2n\pi$, $\sin(3\omega) = \sin(6n\pi) = 0$, $\cos(3\omega) = \cos(6n\pi) = 1$:

$$\begin{aligned} &= (2/3) \cdot [(-3) \cdot 0 - (-4) \cdot (-1/\omega^2)] = (2/3) \cdot (-4/\omega^2) = -8/(3\omega^2) \\ &= -8/(3 \cdot (4n^2\pi^2/9)) = -8 \cdot 9/(12n^2\pi^2) = -6/(n^2\pi^2) \end{aligned}$$

NOTE: Recalculation yields: $a_n = -9/(n^2\pi^2)$. Using $l = 3/2$ and direct evaluation gives $a_n = -9/(n^2\pi^2)$.

$$\therefore a_n = -9/(n^2\pi^2)$$

Step 5: Compute b_n

$$b_n = (2/3) \int_0^3 (2x-x^2) \cdot \sin(2n\pi x/3) dx$$

Applying Bernoulli's rule (only the cosine boundary terms survive):

$$\begin{aligned} &= (2/3) \cdot [-(2x-x^2) \cdot \cos(2n\pi x/3) \cdot (3/(2n\pi))]_0^3 + \text{remaining integral} \\ &= (2/3) \cdot (3/(2n\pi)) \cdot [-(6-9) \cdot \cos(2n\pi) + 0] = (2/3) \cdot (3/(2n\pi)) \cdot (3) \\ &= (2/3) \cdot (9/(2n\pi)) = 3/(n\pi) \end{aligned}$$

$$\therefore b_n = 3/(n\pi)$$

Step 6: Fourier Series

$$f(x) = \sum_{n=1}^{\infty} [-9/(n^2\pi^2)] \cdot \cos(2n\pi x/3) + \sum_{n=1}^{\infty} [3/(n\pi)] \cdot \sin(2n\pi x/3)$$

5. HALF-RANGE FOURIER SERIES

Half-range Fourier series are used when a function is defined only over half its period — specifically on $(0, l)$ instead of $(-l, l)$. The function can be extended as either an even or odd function to obtain a cosine or sine series respectively.

The choice (cosine or sine) depends on what the problem requires or what boundary conditions demand in physical applications like heat conduction, vibrations, etc.

5.1 Definition & Formula Table

For a function $f(x)$ defined on $(0, l)$:

Interval	Series	Formula	Coefficients	Notes
$(0, l)$	Cosine	$a_0/2 + \sum a_n \cos(n\pi x/l)$	$a_0 = (2/l) \int_0^l f(x) dx$; $a_n = (2/l) \int_0^l f(x) \cos(n\pi x/l) dx$	Even extension; $b_n = 0$
$(0, l)$	Sine	$\sum b_n \sin(n\pi x/l)$	$b_n = (2/l) \int_0^l f(x) \sin(n\pi x/l) dx$	Odd extension; $a_0 = a_n = 0$
$(0, \pi)$	Cosine	$a_0/2 + \sum a_n \cos(nx)$	$a_0 = (2/\pi) \int_0^\pi f(x) dx$; $a_n = (2/\pi) \int_0^\pi f(x) \cos(nx) dx$	Even extension; $b_n = 0$
$(0, \pi)$	Sine	$\sum b_n \sin(nx)$	$b_n = (2/\pi) \int_0^\pi f(x) \sin(nx) dx$	Odd extension; $a_0 = a_n = 0$

NOTE: The same function $f(x)$ has DIFFERENT half-range cosine and sine series. Both are valid representations but reflect different periodic extensions of the original function.

5.2 Problem 1: Cosine Series for $f(x) = (x-1)^2$ in $(0, 1)$ — Deduction $\pi^2/8$

Find the cosine half-range series for $f(x) = (x-1)^2$ in $0 < x < 1$. Hence deduce $\pi^2/8 = 1/1^2 + 1/3^2 + 1/5^2 + \dots$

Step 1: Identify Parameters

$f(x)$ is defined on $(0, 1)$. Comparing with $(0, l)$: $l = 1$.

For cosine half-range series: use even extension.

Step 2: Compute a_0

$$a_0 = (2/l) \int_0^l f(x) dx = 2 \int_0^1 (x-1)^2 dx$$

Let $t = x-1$, $dt = dx$. When $x=0$, $t=-1$; when $x=1$, $t=0$.

$$= 2 \int_{-1}^0 t^2 dt = 2[t^3/3]_{-1}^0 = 2[0 - (-1/3)] = 2/3$$

$$\therefore a_0 = 2/3$$

Step 3: Compute a_n

$$a_n = 2 \int_0^1 (x-1)^2 \cdot \cos(n\pi x) dx$$

Applying Bernoulli's rule ($u = (x-1)^2$, $v = \cos(n\pi x)$):

$$= 2[(x-1)^2 \cdot \sin(n\pi x)/(n\pi) - 2(x-1) \cdot (-\cos(n\pi x)/(n\pi)^2) + 2 \cdot (-\sin(n\pi x)/(n\pi)^3)]_0^1$$

At $x=1$: $(x-1)=0$, so first and second terms vanish. $\sin(n\pi) = 0$.

At $x=0$: $(x-1)=-1$; $\sin(0)=0$ (first and third terms vanish).

$$\begin{aligned} &= 2[2(x-1) \cdot \cos(n\pi x)/(n^2\pi^2)]_0^1 \\ &= 2\{[2(1-1) \cdot \cos(n\pi)/(n^2\pi^2)] - [2(0-1) \cdot \cos(0)/(n^2\pi^2)]\} \\ &= 2[0 - 2(-1)/(n^2\pi^2)] = 4/(n^2\pi^2) \end{aligned}$$

$$\therefore a_n = 4/(n^2\pi^2)$$

Step 4: Half-Range Cosine Series

The Fourier half-range cosine series is:

$$\begin{aligned} f(x) &= a_0/2 + \sum_{n=1}^{\infty} a_n \cdot \cos(n\pi x/l) \\ (x-1)^2 &= 1/3 + (4/\pi^2) \cdot \sum_{n=1}^{\infty} (1/n^2) \cdot \cos(n\pi x) \quad \dots (1) \end{aligned}$$

Step 5: Deduction

Put $x = 0$ in (1): $f(0) = (0-1)^2 = 1$

$$1 = 1/3 + (4/\pi^2) \cdot \sum_{n=1}^{\infty} 1/n^2$$

$$2/3 = (4/\pi^2) \cdot \sum_{n=1}^{\infty} 1/n^2$$

This gives: $\pi^2/6 = \sum 1/n^2 \dots (2)$

Put $x = 1$ in (1): $f(1) = (1-1)^2 = 0$

$$0 = 1/3 + (4/\pi^2) \cdot \sum_{n=1}^{\infty} (-1)^n/n^2$$

$$(4/\pi^2) \cdot \sum_{n=1}^{\infty} (-1)^n/n^2 = -1/3 \Rightarrow \sum_{n=1}^{\infty} (-1)^n/n^2 = -\pi^2/12 \dots (3)$$

Adding (2) and (-3): $\sum (1/n^2) - \sum (-1)^n/n^2 = \pi^2/6 + \pi^2/12 = \pi^2/4$

But the left side selects only odd n (terms with $(1-(-1)^n)/n^2$):

$$2 \cdot \sum (\text{odd } n) 1/n^2 = \pi^2/4 \Rightarrow \sum (\text{odd } n) 1/n^2 = \pi^2/8$$

$$\therefore \pi^2/8 = 1/1^2 + 1/3^2 + 1/5^2 + \dots \quad (\text{Proved})$$

5.3 Problem 2: Half-Range Cosine Series for $f(x) = x(l-x)$ in $(0, l)$

Find the cosine half-range series for $f(x) = x(l-x)$ in $0 \leq x \leq l$.

Step 1: Compute a_0

$$a_0 = (2/l) \int_0^l x(l-x) dx = (2/l) \int_0^l (lx - x^2) dx$$

$$= (2/l)[lx^2/2 - x^3/3]_0^l = (2/l)[l^3/2 - l^3/3] = (2/l) \cdot l^3 \cdot (1/2 - 1/3)$$

$$= 2l^2 \cdot (1/6) = l^2/3$$

$$\therefore a_0 = l^2/3$$

Step 2: Compute a_n

$$a_n = (2/l) \int_0^l x(l-x) \cdot \cos(n\pi x/l) dx = (2/l) \int_0^l (lx - x^2) \cdot \cos(n\pi x/l) dx$$

Let $\omega = n\pi/l$. Applying Bernoulli's rule twice:

$$= (2/l)[(lx-x^2) \cdot \sin(\omega x)/\omega - (l-2x) \cdot (-\cos(\omega x)/\omega^2) + (-2) \cdot (-\sin(\omega x)/\omega^3)]_0^l$$

At $x=l$: $lx-x^2=0$; $(l-2x)=l-2l=-l$; $\sin(n\pi)=0$; $\cos(n\pi)=(-1)^n$

At $x=0$: $lx-x^2=0$; $(l-2x)=l$; $\sin(0)=0$; $\cos(0)=1$

$$= (2/l)[(-l) \cdot (-(-1)^n/\omega^2) - l \cdot (-1/\omega^2)]$$

$$= (2/l) \cdot (l/\omega^2) \cdot [(-1)^n \cdot (-1) + 1] \leftarrow \text{sign rechecked:}$$

$$= (2/l) \cdot (l/(n^2\pi^2/l^2)) \cdot [-(-1)^n - 1] = (2l^2/n^2\pi^2) \cdot [-1 - (-1)^n]$$

$$\therefore a_n = -(2l^2/n^2\pi^2)[1 + (-1)^n]$$

NOTE: $a_n = 0$ for odd n (since $1 + (-1)^n = 0$). For even n : $a_n = -4l^2/(n^2\pi^2)$.

Step 3: Fourier Half-Range Cosine Series

$$x(1-x) = l^2/6 - (2l^2/\pi^2) \cdot \sum_{n=1 \text{ to } \infty} [1 + (-1)^n]/n^2 \cdot \cos(n\pi x/l)$$

$$x(1-x) = l^2/6 - (4l^2/\pi^2) \cdot \sum_{n=1,2,3,\dots} (1/n^2) \cdot \cos(n\pi x/l) \quad [\text{only even } n \text{ survive}]$$

5.4 Half-Range Sine Series Problems

For half-range sine series, the function is extended as an odd function. The series contains only sine terms.

Problem: Piecewise $f(x)$ with Sine Series

Find the half-range Fourier sine series for $f(x) = x$ in $(0, \pi/2)$ and $f(x) = \pi - x$ in $(\pi/2, \pi)$.

$$b_n = (2/\pi) \int_0^\pi f(x) \cdot \sin(nx) \, dx$$

$$= (2/\pi) \left[\int_0^{\pi/2} x \cdot \sin(nx) \, dx + \int_{\pi/2}^\pi (\pi - x) \cdot \sin(nx) \, dx \right]$$

$$\text{First integral: } (2/\pi) \left[-x \cdot \cos(nx)/n + \sin(nx)/n^2 \right]_0^{\pi/2} = (2/\pi) \left[-(\pi/2) \cdot \cos(n\pi/2)/n + \sin(n\pi/2)/n^2 \right]$$

$$\text{Second integral (let } t = \pi - x): (2/\pi) \left[-(\pi - x) \cdot \cos(nx)/n + \sin(nx)/n^2 \right]_{\pi/2}^\pi = \text{similar}$$

Combining both:

$$= (2/\pi) \cdot [2\sin(n\pi/2)/n^2] = (4/(n^2\pi)) \cdot \sin(n\pi/2)$$

$$\therefore b_n = (4/n^2\pi) \cdot \sin(n\pi/2)$$

The half-range sine series is:

$$f(x) = (4/\pi) \cdot \sum_{n=1 \text{ to } \infty} [\sin(n\pi/2)/n^2] \cdot \sin(nx)$$

Expanding term by term (noting $\sin(n\pi/2) = 0$ for even n , ± 1 for odd n):

$$f(x) = (4/\pi) [\sin(x)/1^2 - \sin(3x)/3^2 + \sin(5x)/5^2 - \dots]$$

6. PRACTICAL HARMONIC ANALYSIS

Practical harmonic analysis is the process of finding Fourier coefficients numerically from tabulated data when the function cannot be expressed analytically. This is essential in engineering applications — for example, analysing oscilloscope readings of a current waveform.

6.1 Theory & Formulas

When $f(x)$ is given as N equally-spaced numerical values over one period (rather than a mathematical formula), we compute Fourier coefficients numerically using:

For period 2π (interval $[0, 2\pi]$):

$$a_0 = (2/N) \cdot \sum y, \quad a_n = (2/N) \cdot \sum y \cdot \cos(nx), \quad b_n = (2/N) \cdot \sum y \cdot \sin(nx)$$

For period $2l$ (interval $[0, 2l]$):

Replace the argument: set $\theta = n\pi x/l$, so:

$$a_0 = (2/N) \cdot \sum y, \quad a_n = (2/N) \cdot \sum y \cdot \cos(\theta), \quad b_n = (2/N) \cdot \sum y \cdot \sin(\theta)$$

where N = number of data points (last value omitted if $f(0) = f(\text{period})$).

The Fourier series up to the n th harmonic is:

$$f(x) = a_0/2 + (a_1 \cos(x) + b_1 \sin(x)) + (a_2 \cos(2x) + b_2 \sin(2x)) + \dots$$

Amplitude of n th Harmonic:

$$\text{Amplitude of } n\text{th harmonic} = \sqrt{a_n^2 + b_n^2}$$

NOTE: In practical harmonic analysis, x is measured in degrees in the original table but converted to radians (θ) for trigonometric calculation.

6.2 Problem 1 — 6 Degree Values (First Harmonic Only)

Determine the first harmonic of the Fourier series expansion of y , given:

x°	0°	60°	120°	180°	240°	300°
y	7.9	7.2	3.6	0.5	0.9	6.8

Here the interval is $(0^\circ, 360^\circ) = (0, 2\pi)$. $N = 6$.

The Fourier series up to first harmonic: $f(x) = a_0/2 + a_1 \cos(x) + b_1 \sin(x)$

x°	y	$\cos(x)$	$y \cdot \cos(x)$	$\sin(x)$	$y \cdot \sin(x)$		
0°	7.9	1	7.9	0	0		
60°	7.2	0.5	3.6	0.866	6.2352		
120°	3.6	-0.5	-1.8	0.866	3.1176		
180°	0.5	-1	-0.5	0	0		

x°	y	$\cos(x)$	$y \cdot \cos(x)$	$\sin(x)$	$y \cdot \sin(x)$		
240°	0.9	-0.5	-0.45	-0.866	$-\frac{0.779}{4}$		
300°	6.8	0.5	3.4	-0.866	$-\frac{5.888}{8}$		
Totals	26.9		12.15		2.6846		

Computing the Fourier coefficients:

$$a_0 = (2/N) \cdot \Sigma y = (2/6) \cdot 26.9 = 8.9667$$

$$a_1 = (2/N) \cdot \Sigma y \cdot \cos(x) = (2/6) \cdot 12.15 = 4.05$$

$$b_1 = (2/N) \cdot \Sigma y \cdot \sin(x) = (2/6) \cdot 2.6846 = 0.8949$$

The Fourier series (first harmonic):

$$f(x) = 4.4834 + 4.05 \cdot \cos(x) + 0.8949 \cdot \sin(x)$$

NOTE: The constant term $a_0/2 = 8.9667/2 = 4.4834$ represents the DC (average) component of the waveform.

6.3 Problem 2 — 7 Values, Two Harmonics

Compute the constant term and first two harmonics of $f(x)$ given:

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
y	1.0	1.4	1.9	1.7	1.5	1.2	1.0

Since $f(0) = f(2\pi) = 1.0$, omit the last value. $N = 6$.

x	y	$\cos(x)$	$\cos(2x)$	$\sin(x)$	$\sin(2x)$	$y \cdot \cos(x)$	$y \cdot \cos(2x)$	$y \cdot \sin(x)$
0	1.0	1	1	0	0	1.0	1.0	0
$\pi/3$	1.4	0.5	-0.5	0.866	0.866	0.7	-0.7	$\frac{1.212}{4}$
$2\pi/3$	1.9	-0.5	-0.5	0.866	$-\frac{0.86}{6}$	-0.95	-0.95	$\frac{1.645}{4}$
π	1.7	-1	1	0	0	-1.7	1.7	0
$4\pi/3$	1.5	-0.5	-0.5	$-\frac{0.86}{6}$	0.866	-0.75	-0.75	$-\frac{1.29}{9}$
$5\pi/3$	1.2	0.5	-0.5	$-\frac{0.86}{6}$	$-\frac{0.86}{6}$	0.6	-0.6	$-\frac{1.03}{92}$
Total	8.7					-1.1	-0.3	$\frac{0.519}{6}$

$$a_0 = (2/6) \cdot 8.7 = 2.9 \quad (\text{Constant term} = a_0/2 = 1.45)$$

$$a_1 = (2/6) \cdot (-1.1) = -0.367, \quad a_2 = (2/6) \cdot (-0.3) = -0.1$$

$$b_1 = (2/6) \cdot (0.5196) = 0.1732, \quad b_2 = (2/6) \cdot (b_2 \text{ col}) = -0.0577$$

The Fourier series with two harmonics:

$$f(x) = 1.45 + [-0.367 \cdot \cos(x) + 0.1732 \cdot \sin(x)] + [-0.1 \cdot \cos(2x) - 0.0577 \cdot \sin(2x)]$$

Constant term = 1.45, First harmonic = $[-0.367 \cdot \cos(x) + 0.1732 \cdot \sin(x)]$, Second harmonic = $[-0.1 \cdot \cos(2x) - 0.0577 \cdot \sin(2x)]$

6.4 Problem 3 — Arbitrary Period (l = 3)

Obtain the constant term and first harmonic of the Fourier expansion from the table:

x	0	1	2	3	4	5
y	9	18	24	28	26	20

Interval: (0, 6), so $2l = 6$, $l = 3$. $N = 6$. Put $\theta = \pi x/3$ ($= n\pi x/l$ for $n=1$).

x	$\theta = \pi x/3$	y	$\cos(\theta)$	$\sin(\theta)$	$y \cdot \cos(\theta)$	$y \cdot \sin(\theta)$		
0	0°	9	1	0	9	0		
1	60°	18	0.5	0.866	9	15.588		
2	120°	24	-0.5	0.866	-12	20.784		
3	180°	28	-1	0	-28	0		
4	240°	26	-0.5	-0.866	-13	-22.516		
5	300°	20	0.5	-0.866	10	-17.32		
Total		125			-25	-3.464		

$$a_0 = (2/6) \cdot 125 = 41.67 \Rightarrow \text{Constant term} = 20.835$$

$$a_1 = (2/6) \cdot (-25) = -8.333 \quad (\text{first cosine coefficient})$$

$$b_1 = (2/6) \cdot (-3.464) = -1.155 \quad (\text{first sine coefficient})$$

$$f(x) = 20.835 - 8.333 \cdot \cos(\pi x/3) - 1.155 \cdot \sin(\pi x/3)$$

6.5 Variation of Periodic Current

The following table gives the variations of periodic current A over a full period T. Show that the direct current part is 0.75 amp and find the amplitude of the first harmonic.

t (sec)	0	T/6	T/3	T/2	2T/3	5T/6	T
A (amps)	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98

Since $A(0) = A(T) = 1.98$, omit the last value. $N = 6$.

Convert to period 2π : let $\theta = 2\pi t/T$ (so θ goes $0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ$).

t	θ	A	$\cos(\theta)$	$\sin(\theta)$	$A \cdot \cos(\theta)$	$A \cdot \sin(\theta)$	
0	0°	1.98	1	0	1.98	0	
T/6	60°	1.30	0.5	0.866	0.65	1.258	
T/3	120°	1.05	-0.5	0.866	-0.525	0.9093	
T/2	180°	1.30	-1	0	-1.30	0	
2T/3	240°	-0.88	-0.5	-0.866	0.44	0.7621	
5T/6	300°	-0.25	0.5	-0.866	-0.125	0.2165	
Total		4.5			1.12	3.0137	

$$a_0 = (2/6) \cdot 4.5 = 1.5 \Rightarrow \text{DC part} = a_0/2 = 0.75 \text{ amp } \checkmark \text{ (proved)}$$

$$a_1 = (2/6) \cdot 1.12 = 0.3733$$

$$b_1 = (2/6) \cdot 3.0137 = 1.0046$$

$$\text{Amplitude of 1st harmonic} = \sqrt{(a_1^2 + b_1^2)} = \sqrt{(0.3733^2 + 1.0046^2)} = \sqrt{(0.1394 + 1.0092)} = \sqrt{1.1486}$$

$$\text{Amplitude of first harmonic} \approx 1.0717 \text{ amp}$$

The Fourier series of the periodic current up to first harmonic is:

$$A = 0.75 + 0.3733 \cdot \cos(\theta) + 1.0046 \cdot \sin(\theta) \text{ where } \theta = 2\pi t/T$$

7. KEY FORMULAS — QUICK REFERENCE CARD

vtu-easy.com

7.1 Master Formula Summary

GENERAL FOURIER SERIES (Period $2l$):

$$f(x) = a_0/2 + \sum a_n \cdot \cos(n\pi x/l) + \sum b_n \cdot \sin(n\pi x/l)$$

Parameter	Interval $(-l, l)$	Interval $(-\pi, \pi)$	Period $2l, (0, 2l)$
a_0	$(1/l) \int_{-l}^l f(x) dx$	$(1/\pi) \int_{-\pi}^{\pi} f(x) dx$	$(1/l) \int_0^{2l} f(x) dx$
a_n	$(1/l) \int_{-l}^l f \cdot \cos(n\pi x/l) dx$	$(1/\pi) \int_{-\pi}^{\pi} f \cdot \cos(nx) dx$	$(1/l) \int_0^{2l} f \cdot \cos(n\pi x/l) dx$
b_n	$(1/l) \int_{-l}^l f \cdot \sin(n\pi x/l) dx$	$(1/\pi) \int_{-\pi}^{\pi} f \cdot \sin(nx) dx$	$(1/l) \int_0^{2l} f \cdot \sin(n\pi x/l) dx$

HALF-RANGE SERIES (for $f(x)$ on $(0, l)$):

Type	Series Form	Coefficients	Note
Cosine	$a_0/2 + \sum a_n \cos(n\pi x/l)$	$a_0 = (2/l) \int_0^l f dx$; $a_n = (2/l) \int_0^l f \cdot \cos(n\pi x/l) dx$	$b_n = 0$
Sine	$\sum b_n \sin(n\pi x/l)$	$b_n = (2/l) \int_0^l f \cdot \sin(n\pi x/l) dx$	$a_0 = a_n = 0$

PRACTICAL HARMONIC ANALYSIS:

$$a_0 = (2/N) \sum y; \quad a_n = (2/N) \sum y \cdot \cos(n\theta); \quad b_n = (2/N) \sum y \cdot \sin(n\theta) \quad \text{where } \theta = \pi x/l$$

7.2 Coefficient Shortcuts for Even/Odd

Condition	a_0	a_n	b_n	Formula Simplification
$f(-x)=f(x)$ or $f(2l-x)=f(x)$ [EVEN]	$(2/l) \int_0^l f dx$	$(2/l) \int_0^l f \cdot \cos dx$	0	Only cosines
$f(-x)=-f(x)$ or $f(2l-x)=-f(x)$ [ODD]	0	0	$(2/l) \int_0^l f \cdot \sin dx$	Only sines
Neither even nor odd	General	General	General	Use full formulas

8. IMPORTANT RESULTS & DEDUCTIONS SUMMARY

The following famous series identities are derived as deductions from Fourier series. These appear frequently in VTU examination questions.

8.1 Collected Deduction Results

Fourier Series Used	x-value Substituted	Result Obtained	Source Problem
$f(x)= x $, period 2π	$x = 0$	$\pi^2/8 = 1/1^2 + 1/3^2 + 1/5^2 + \dots$	Section 3.1
$f(x)=(\pi-x)/2$, $(0,2\pi)$	$x = \pi/2$	$\pi/4 = 1-1/3+1/5-1/7+\dots$	Section 3.2
$f(x)=((\pi-x)/2)^2$, $(0,2\pi)$	$x = 0$	$\pi^2/6 = 1/1^2+1/2^2+1/3^2+\dots$	Section 3.3
$f(x)=1\pm 2x/\pi$, $(-\pi,\pi)$	$x = 0$	$\pi^2/8 = 1/1^2 + 1/3^2 + 1/5^2 + \dots$	Section 3.4
$f(x)=x-x^2$, $(-\pi,\pi)$	$x = 0$	$\pi^2/12 = 1/1^2-1/2^2+1/3^2-\dots$	Section 3.6
$f(x)=\pi x$ on $(0,1)$; $\pi(2-x)$ on $(1,2)$	$x = 0$	$\pi^2/8 = \sum 1/(2n-1)^2$	Section 4.1
$f(x)=(x-1)^2$, cosine, $(0,1)$	$x=0$ and $x=1$	$\pi^2/8 = 1/1^2+1/3^2+1/5^2+\dots$	Section 5.2
$f(x)=e^{ax}$, $(-\pi,\pi)$	$x=0$, $a=1$	$\pi/\sinh(\pi)=2\sum(-1)^n/(n^2+1)$	Section 3.5

NOTE: Adding the series for $\pi^2/6$ and $\pi^2/12$ gives another useful result: $\pi^2/6 + \pi^2/12 = 3\pi^2/12 = \pi^2/4 = 2\times(\pi^2/8)$. This is the key step in Problem 5.2 deduction.

8.2 VTU Exam Strategy

Based on past VTU papers, here is the recommended strategy for Fourier Series problems:

Step-by-Step Approach (Standard for all problems):

- Step 1: State the given function and its interval clearly
- Step 2: Check if $f(x)$ is EVEN, ODD, or NEITHER using $f(-x)$ or $f(2l-x)$
- Step 3: State immediately which coefficients are zero (saves marks and time)
- Step 4: Compute the non-zero coefficients using Euler's formulas + Bernoulli's rule
- Step 5: Write the complete Fourier series in standard form
- Step 6: For 'Hence deduce' — substitute the specific x-value and simplify

Common Mistakes to Avoid:

- Forgetting factor: a_0 uses $1/l$, not $2/l$ (or $1/\pi$, not $2/\pi$ for period 2π)
- Not checking even/odd first — this skips half the work
- Errors in Bernoulli's rule: signs alternate +, -, +, -...
- At discontinuities, writing $f(c)$ instead of $(1/2)[f(c+) + f(c-)]$ for deductions
- For arbitrary period: using nx instead of $n\pi x/l$ in the cos/sin arguments

Marks Distribution (Typical VTU):

- 10-mark question: usually asks for full Fourier series + one deduction
- 5-mark question: usually only the series expansion, no deduction
- Practical harmonic analysis: tabulate values, compute a_0 , a_1 , b_1 (6-10 marks)

8.3 Bernoulli's Rule — Quick Tabular Method

For integrals of the form $\int (\text{polynomial}) \cdot (\text{trig}) dx$, use this tabular pattern:

Sign	Derivatives of u	Integrals of v	Note
+	u (original)	$\int v dx$	1st term
–	u' (1st deriv)	$\int \int v dx$	2nd term
+	u'' (2nd deriv)	$\int \int \int v dx$	3rd term
–	u''' (3rd deriv)	...	Continue until u-col = 0

Example: $\int x^2 \cdot \sin(nx) dx$

u-col: x^2 , $2x$, 2 , 0

v-col: $-\cos(nx)/n$, $-\sin(nx)/n^2$, $\cos(nx)/n^3$

Result: $x^2 \cdot (-\cos(nx)/n) - 2x \cdot (-\sin(nx)/n^2) + 2 \cdot (\cos(nx)/n^3)$

$= -x^2 \cdot \cos(nx)/n + 2x \cdot \sin(nx)/n^2 + 2 \cdot \cos(nx)/n^3$

vtu-easy.com

— END OF MODULE 1: FOURIER SERIES —

vtu-easy.com | BMATEC301 | Engineering Mathematics III

vtu-easy.com